Signal Processing and Social Signal Processing

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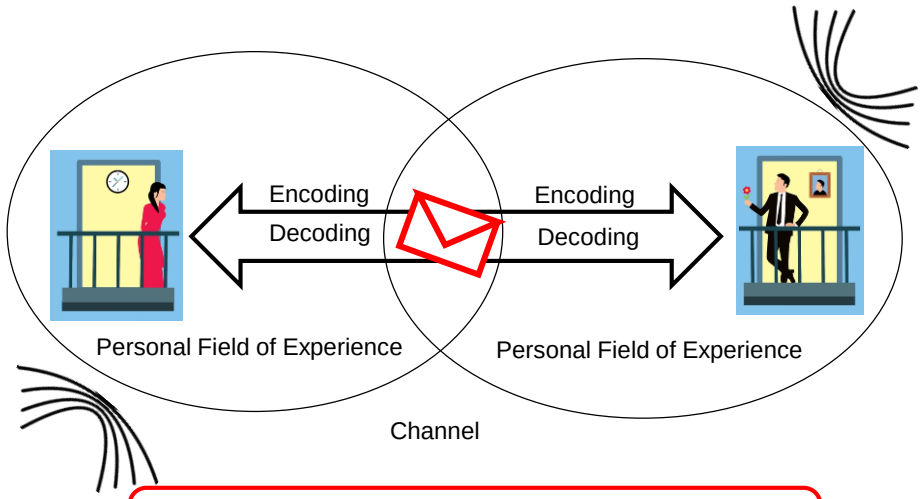
Let's go back to Romeo and Juliet



They exchange messages, but technically we should say signals…

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Signals & Co.

- What is a signal?
  - Any measurable quantity that conveys information
  - Any physical quantity that varies (i.e. a function) with time / space, and / or another independent variable
  - Examples:
    - electrical: voltage output of amplifier
    - mechanical: acceleration of a car
    - acoustic: air pressure measured by a microphone
    - biological: body temperature
    - image and video: intensities of each pixel

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Signals & Co.

- Looking at the independent variable(s):
  - Monodimensional vs. Multidimensional signals
    - A monodimensional signal is a function of a single independent variable (e.g. amplitude of a sound over time);
    - A multidimensional signal depends on 2 / more independent variables. (e.g. an image is a 2D signal, as its pixel intensity is a function of spatial coordinates (x,y). A video is a 3D signal, as it depends on spatial coordinates and time (x,y,t)
  - If the independent variable is continuous ($\in$ R), the signal is called a continuous signal
  - If the independent variable is discrete ($\in$ N or $\in$ Z), the signal is called a discrete signal

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Signals & Co.

- Looking at the dependent variable(s), that is at the signal amplitude:
  - Monochannel vs. Multichannels signals
    - A monochannel signal has a single output (e.g. a monophonic audio recording);
    - A multichannel signal is composed of multiple independent output (e.g. stereo audio recording; a color image)
  - If the dependent variable is continuous ($\in$ R), the signal has a continuous amplitude
  - If the dependent variable is discrete ($\in$ N or $\in$ Z), the signal has a discrete amplitude

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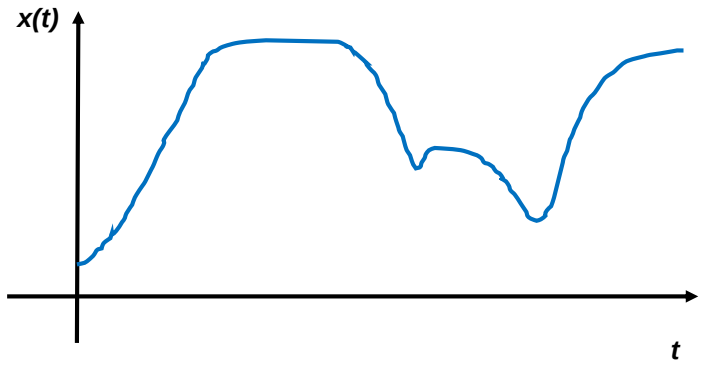
Signals & Co.

- A continuous-time signal with continuous amplitude is called an analog signal
- E.g.



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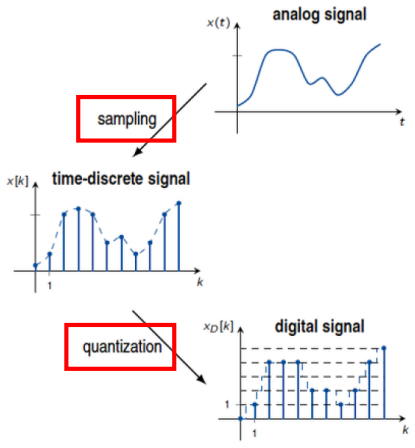
First Operations on Signals



A digital signal is a discrete-time signal having a discrete amplitude

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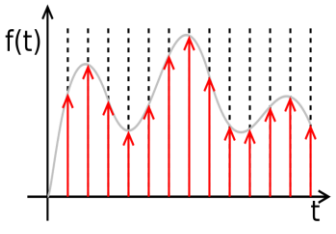
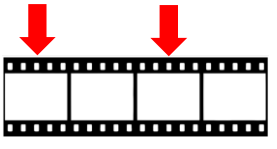
Sampling and quantization

- Sampling is the reduction of a continuous signal to a discrete signal. A sample is a value at a point in time.





- The distance between two samples is called sampling period, its reciprocal is called sampling frequency
- At which frequency I have to sample a signal to avoid to loose information?

  The Nyquist theorem says that: The sampling frequency must be at least the Nyquist rate, which is defined as twice the highest frequency component of the signal

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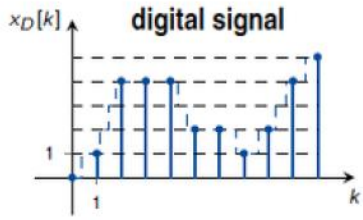
Quantization

- Quantization is the process of approximating (mapping) a continuous range of values by a relatively small (finite) set of discrete symbols or integer values.



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Signals Processing vs. Digital Signal Processing

- Signal processing is a field of electrical engineering that focuses on the analysis, manipulation, and synthesis of signals
- Digital signal processing (DSP) is a subfield of signal processing that deals exclusively with digital signals

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Operating on signals

- Transforms
  Sometimes, it's appropriate to change the domain of observation of a signal to highlight properties that are invisible in the original domain (e.g. time)

  The most well-known is the Fourier Transform:
  any integrable signal can be expressed as a sum of sinusoidal functions. The new domain is the frequency domain

$$F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-j\omega t} dt$$

Fourier Transform

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j\frac{2\pi}{N}kn}$$

Discrete Fourier Transform

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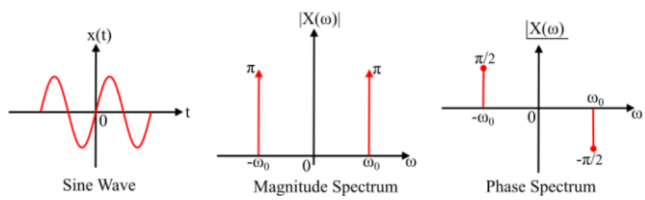
Operating on signals

- Transforms
  The result of applying the Fourier Transform is a complex number that can be represented as magnitude and phase:

  - The magnitude represents the amplitude of each sinusoidal function

  - The phase describes the starting position of each sinusoidal function



source: tutorialspoint

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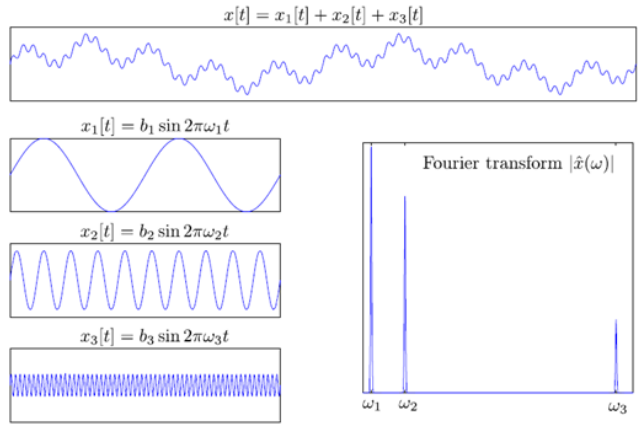
A more complex example



source: adapted from Donzé et al. 2012

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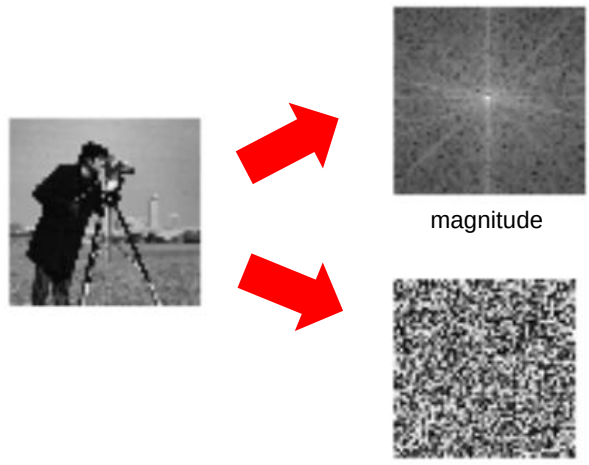
Examples



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Operating on signals

- Filtering: extracting desired information from a signal by removing unwanted components
- Three kinds of filters:
  - *Low-pass:* they allow frequencies below a specific threshold (the cutoff frequency) to pass through while attenuating, or reducing, frequencies above it. Particularly useful for removing high-frequency noise from a signal
  - *High-pass*: They allow frequencies above the cutoff frequency to pass through while attenuating lower frequencies. For example, they are used to remove DC or low-frequency components from a signal
  - *Band-pass*: They allow the passage of frequencies within a specific range. They are useful for isolating a frequency band of interest
- From a technically point of view, filtering is implemented as a convolution in the original domain between the signal and the filter function

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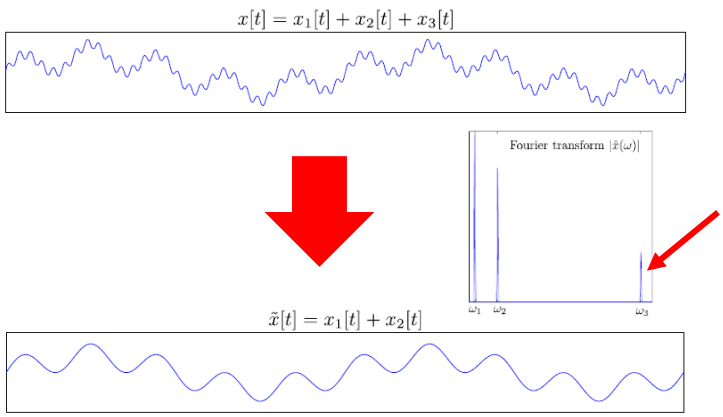
Simple Example

- Removing the highest frequency:



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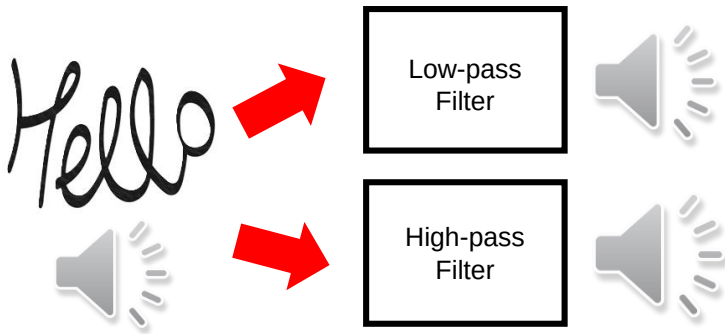
Examples



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Examples

Low-pass Filter

High-pass Filter

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Operating on signals

- Convolution:
  It's a mathematical operation that combines two functions to produce a third one, which expresses how the shape of one function is modified by the other

  In simpler terms, convolution is an operation that "merges" two signals, a main one and a kernel that serves to modify the first

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Operating on signals

- Convolution over time:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau$$

$$(x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Try it!

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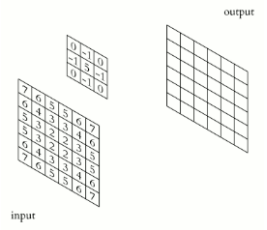
Operating on signals

- Convolution over 2D space:

$$(A * B)[i,j] = \sum_k \sum_l A[k,l]B[i-k,j-l]$$



Try it!

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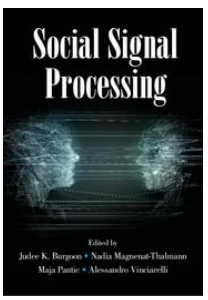
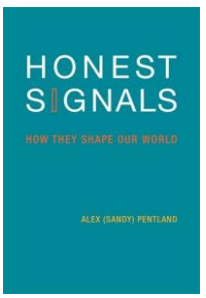
Social Signal Processing (SSP)

- Basically, it is signal processing applied to humans
- Formal defintion: *"SSP is the computing domain aimed at modeling, analysis, and synthesis of social signals in human–human and human–machine interactions"*
(Pentland, 2007; Vinciarelli et al., 2008, 2012; Vinciarelli, Pantic, & Bourlard, 2009)





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Social Signal Processing (SSP)

- …applied to humans: facial expression, gesture, voice, gaze and so on

- "social signals are machine detectable, *"communicative or informative signals which, either directly or indirectly, provides provide information about social facts"*

(Poggi & D'Errico, 2012)

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Quiz time! What is going on?

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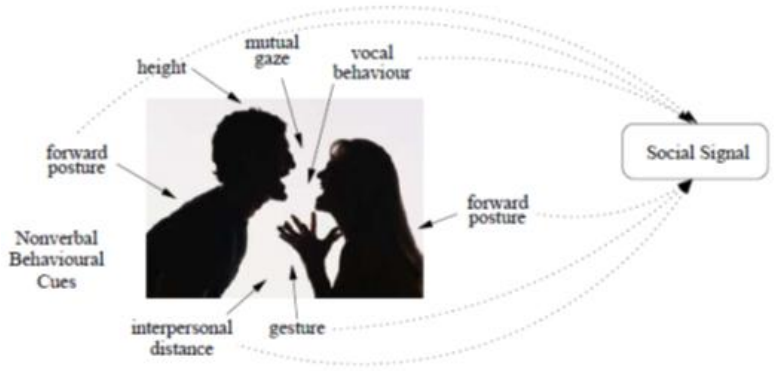
A closer look



Picture from (Vinciarelli, Pantic, Boulard, 2009)

- Verbal and nonverbal behavioural cues produce or are combined (multimodality!) to produce social signals
- Verbal cues account for just 7% of the overall social perception (Knapp & Hall, 1972)

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